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INTERACTIVE COURSE

Introduction to Regression with statsmodels in Python

Intermediate | Updated: March 2026

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4 hr | 14 videos | 53 Exercises | 58,617 | 4150 XP

What you'll learn

Description

Linear regression and logistic regression are two of the most widely used statistical models. They act like master keys, unlocking the secrets hidden in your data. In this course, you'll gain the skills you need to fit simple linear and logistic regressions. Through hands-on exercises, you'll explore the relationships between variables in real-world datasets, including motor insurance claims, Taiwan house prices, fish sizes, and more. By the end of this course, you'll know how to make predictions from your data, quantify model performance, and diagnose problems with model fit.

The videos contain live transcripts you can reveal by clicking "Show transcript" at the bottom left of the videos.
The course glossary can be found on the right in the resources section.
To obtain CPE credits you need to complete the course and reach a score of 70% on the qualified assessment. You can navigate to the assessment by clicking on the CPE credits callout on the right.

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PREREQUISITES
Introduction to Data Visualization with Seaborn
Introduction to Statistics in Python

RESOURCES

1 Simple Linear Regression Modeling

100%

You'll learn the basics of this popular statistical model, what regression is, and how linear and logistic regressions differ. You'll then learn how to fit simple linear regression models with numeric and categorical explanatory variables, and how to describe the relationship between the response and explanatory variables using model coefficients.

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Practice Chapter

2 Predictions and model objects

100%

In this chapter, you'll discover how to use linear regression models to make predictions on Taiwanese house prices and Facebook advert clicks. You'll also grow your regression skills as you get hands-on with model objects, understand the concept of "regression to the mean", and learn how to transform variables in a dataset.

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Practice Chapter

3 Assessing model fit

100%

In this chapter, you'll learn how to ask questions of your model to assess fit. You'll learn how to quantify how well a linear regression model fits, diagnose model problems using visualizations, and understand each observation's leverage and influence to create the model.

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Practice Chapter

Create Course Notes
Your learnings in one place

Customer churn data
Dataset

Taiwan real estate...
Dataset

Ad conversion data
Dataset

S&P 500 data
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Fish measurement d...
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Course Glossary
Dataset

PART OF THESE TRACKS

Associate Data Scientist

Statistics Fundamentals

INSTRUCTORS

Maarten Van den Broeck